

... would help determine which ones achieved early sales and, over time, which firms dominated”.

Mathematics Inside and Outside the Real World

A mathematical approach to the problem is in order. While in conventional models (such as the well-known Brownian random walk used in finance) the probability of success does not change with every incremental step, only the accumulated wealth, Arthur suggests models such as the Polya process, which is mathematically very difficult to work with, but can be easily understood with the aid of a Monte Carlo simulator. The Polya process can be presented as follows: assume an urn initially containing equal quantities of black and red balls. You are to guess each time which color you will pull out before you make the draw. Here the game is rigged. Unlike a conventional urn, the probability of guessing correctly depends on past success, as you get better or worse at guessing depending on past performance. Thus the probability of winning increases after past wins, that of losing increases after past losses. Simulating such a process, one can see a huge variance of outcomes, with astonishing successes and large number of failures (what we called skewness).

Compare such a process with those that are more commonly modeled, that is, an urn from which the player makes guesses with replacement. Say you played roulette and won. Would this increase your chances of winning again? No. In a Polya process case, it does. Why is this so mathematically hard to work with? Because the notion of independence (i.e., when the next draw does not depend on past outcomes) is violated. Independence is a requirement for working with the (known) math of probability.

What has gone wrong with the development of economics as a science? Answer: there was a bunch of intelligent people who felt compelled to use mathematics just to tell themselves that they were rigorous in their thinking, that theirs was a science. Someone in a great rush decided to introduce mathematical modeling techniques (culprits:

Leon Walras, Gerard Debreu, Paul Samuelson) without considering the fact that either the class of mathematics they were using was too restrictive for the class of problems they were dealing with, or that perhaps they should be aware that the precision of the language of mathematics could lead people to believe that they had solutions when in fact they had none (recall Popper and the costs of taking science too seriously). Indeed the mathematics they dealt with did not work in the real world, possibly because we needed richer classes of processes – and they refused to accept the fact that no mathematics at all was probably better.

The so-called *complexity theorists* came to the rescue. Much excitement was generated by the works of scientists who specialized in nonlinear quantitative methods – the Mecca of those being the Santa Fe Institute near Santa Fe, New Mexico. Clearly these scientists are trying hard, and providing us with wonderful solutions in the physical sciences and better models in the social siblings (though nothing satisfactory there yet). And if they ultimately do not succeed, it will simply be because mathematics may be of only secondary help in our real world. Note another advantage of Monte Carlo simulations is that we can get results where mathematics fails us and can be of no help. In freeing us from equations it frees us from the traps of inferior mathematics. As I said in Chapter 4, mathematics is merely a way of thinking and meditating, little more, in our world of randomness.

BURIDAN'S DONKEY OR THE GOOD SIDE OF RANDOMNESS

Nonlinearity in random outcomes is sometimes used as a tool to break stalemates. Consider the problem of the nonlinear nudge. Imagine a donkey equally hungry and thirsty placed at exactly equal distance from two sources of food and water. In such a framework, he would die of both thirst and hunger as he would be unable to decide which one to get to first. Now inject some randomness in the picture, by randomly nudging the donkey, causing him to get closer to one source, no matter which, and accordingly away from the other. The impasse would be instantly broken and our happy donkey will be either in turn well fed then well hydrated, or well hydrated then well fed.